

Options Intelligence Dashboard Γ

[jones-agent v4](#) — Feature Handbook

JonesTrading · May 2026

Data: Polygon.io | Deploy: Netlify | 112 Tickers

Revised edition — adds Levered ETF Rebalance tab, image-paste OCR data input, historical rebalance capture, and signal rankings update.

Table of Contents

Part I: Dashboard Features

1. Dashboard Overview — Header, market status, ticker search
2. Ticker Detail Card — The main analysis view
3. Header Bar — Ticker, spot price, regime, 25Δ RR badge, skew curve badge
4. Verdict & Conviction Score — Directional positioning assessment
5. Skew Signals Strip — Table 5.4 signal classification
6. Summary Metrics Bar — Dealer flow, Flip/HVL, Squeeze, Δ P/C, NDF
7. GEX Levels Row — Net GEX, Call Wall, Put Wall, Max Pain
8. Volatility Skew Row — 25Δ Skew, Smile Ratio, Risk Reversal, Term Structure
9. Skew Term Structure Row — Near-term, Long-term, Spread
10. Flow Breakdown — Total, 0DTE, Open positioning
11. Options Volume Row — % ADV, 0DTE %, Dealer Hedge
12. Signal Breakdown — Factor-by-factor scoring detail
13. High-Delta Sweeps — Large-print options trades

Part II: Tab Views

14. Scanner Tab — Full watchlist overview table
15. Options Volume Tab — Ranked by options % of ADV
16. Vol Surge Tab — Unusual volume detection
17. Walls Tab — Put and Call wall proximity screen
18. Vol Divergence Tab — Price/vol disagreement scanner
19. Cross-Asset Tab — ETF gamma regimes, squeeze scores, P/C rankings
20. **Levered ETF Rebalance Tab — Mechanical flow estimator (NEW)**

Part III: Reference

21. Interactive Features — Clickable metrics, time series, CSV export
22. Metric Reference — Every metric defined
23. Signal Effectiveness Rankings — 1-10 scoring of all signals

1. Dashboard Overview

The dashboard is a single-page React application that loads snapshot data for 112 tickers captured throughout the trading day. It has two modes of operation:

Snapshot mode (default): Loads the most recent batch capture from Netlify Blobs. This is what you see when the market is closed — the last captured readings from the prior session.

Live lookup mode: When you type a ticker and press Enter, it fetches real-time data directly from Polygon.io — live chain data, live spot price. This works anytime, even after hours.

Header Elements

- **Options Intelligence** — the app title, top left.
- **Market status indicator** — top right. Shows LIVE (green dot) during market hours 9:30-4:00 ET, PRE-MKT (yellow) 9:00-9:30, POST-MKT after close, or CLOSED on weekends.
- **Ticker search bar** — type any ticker symbol. Autocomplete suggests from the 112 known tickers. Press Enter or click Analyze. Recent lookups appear as quick-access buttons.

Auto-Refresh

During market hours, the snapshot reloads every 15 minutes automatically. The batch capture function runs on a Netlify scheduled function and processes all 112 tickers in 4 waves. A countdown timer badge shows time until next refresh. Contextual status badges show: LIVE, PRE-MKT, EOD HOLD, CLOSED.

Tab Bar

Seven tabs organize the dashboard views: Scanner, Opts Volume, Vol Surge, Walls, Vol Divergence, Cross-Asset, and **Rebalance** (new in v4). Each tab provides a different lens on the options positioning data.

2-13. Ticker Detail Card & Components

Chapters 2-13 document the ticker detail card and its component sections. These are unchanged from the v2 handbook. The key components are:

- **Header Bar** (Ch. 3) — Ticker, spot, regime badge, 25 Δ RR badge, skew curve badge
- **Verdict & Conviction Score** (Ch. 4) — Directional assessment with -100 to +100 scoring
- **Skew Signals Strip** (Ch. 5) — Table 5.4 classification: steepening, flattening, inverted, spike, collapsed
- **Summary Metrics Bar** (Ch. 6) — Dealer Flow/1%, Flip/HVL, Squeeze, Δ P/C, NDF
- **GEX Levels Row** (Ch. 7) — Net GEX, Call Wall, Put Wall, Max Pain
- **Volatility Skew Row** (Ch. 8) — 25 Δ Skew, Smile Ratio, Risk Reversal, Term Structure
- **Skew Term Structure Row** (Ch. 9) — Near-term, Long-term, Spread
- **Flow Breakdown** (Ch. 10) — Total, 0DTE, Open positioning with P/C and NDS
- **Options Volume Row** (Ch. 11) — % ADV, 0DTE %, Dealer Hedge
- **Signal Breakdown** (Ch. 12) — Factor-by-factor scoring detail table
- **High-Delta Sweeps** (Ch. 13) — Large-print options trades with dollar size and delta

For complete documentation of chapters 2-13, refer to the v2 handbook. All metrics and computations are unchanged in v4.

14-19. Tab Views (Scanner through Cross-Asset)

Chapters 14-19 document the six original tab views. These are functionally unchanged from v2:

- **Scanner Tab** (Ch. 14) — Full watchlist overview table. Every ticker on one screen with regime, conviction, walls, squeeze, and vol metrics. Sortable columns, click any row to open the detail card.
- **Options Volume Tab** (Ch. 15) — Tickers ranked by options volume as % of average daily volume. Surfaces unusual activity.
- **Vol Surge Tab** (Ch. 16) — Detects tickers with options volume exceeding 2x their trailing average. Attention flag for deeper analysis.
- **Walls Tab** (Ch. 17) — Two-column put/call wall layout. Screens for tickers within 2.5% of a wall holding 20%+ of OI concentration. SPY and QQQ always pinned at top.
- **Vol Divergence Tab** (Ch. 18) — Three signal types: OTM divergence, spot-up/vol-up (speculative call chasing), spot-down/vol-down (IV compression into weakness). Distinct color coding.
- **Cross-Asset Tab** (Ch. 19) — Per-ETF gamma regime breakdown, top squeeze scores, most bullish/bearish P/C names.

For complete documentation of chapters 14-19, refer to the v2 handbook.

20. Levered ETF Rebalance Tab

The Rebalance tab estimates the mechanical equity rebalance flow that levered ETF swap counterparties (banks) must execute daily to maintain their target leverage ratios. This flow is **non-discretionary** — it must happen regardless of price — and it is **always in the direction of the underlying move**: buying on up days, selling on down days. On large move days (2%+), these flows can reach \$5-10B+ across all underlyings combined.

20.1 The Formula

The core rebalance formula is:

$$\text{Rebalance} = X(X-1) \times \Delta S \times \text{AUM}$$

Where:

- **X** = leverage factor (e.g. 3 for TQQQ, -3 for SQQQ)
- **X(X-1)** = the fund's gamma (γ) — the rebalance multiplier. This determines how much rebalance flow a fund generates per 1% move in the underlying, relative to its AUM.
- **ΔS** = today's percentage change in the underlying index
- **AUM** = the fund's net asset value (not leveraged notional)

The key insight is that **gamma grows quadratically with leverage**, and inverse ETFs have disproportionately large gammas:

Leverage	$\gamma = X(X-1)$	Rebalance per 1% move per \$1B AUM	Note
+3x long	6	\$60M	e.g. TQQQ, UPRO, SPXL
+2x long	2	\$20M	e.g. QLD, SSO
-1x inverse	2	\$20M	Same as 2x long!
-2x inverse	6	\$60M	Same as 3x long
-3x inverse	12	\$120M	2x a 3x long — most destabilizing
+1.5x	0.75	\$7.5M	e.g. UVXY

A -3x ETF like SQQQ rebalances **twice as aggressively** as TQQQ (3x long) for the same AUM and same underlying move. This asymmetry means even relatively small inverse ETFs can generate outsized flow.

20.2 Why It Matters: The Destabilization Mechanism

Levered ETF rebalance is **always destabilizing**. On up days, the rebalance is a buy (pushing prices higher). On down days, it is a sell (pushing prices lower). This is true for both long and inverse products — the gamma term $X(X-1)$ is always positive regardless of the sign of X .

The mechanism: the bank maintaining the leverage swap holds a delta of X times the fund's NAV. When the underlying moves by ΔS , the delta drifts and must be adjusted by $\gamma \times \Delta S \times \text{AUM}$ to restore the target leverage ratio. This adjustment trades in the direction of the move because the bank needs more delta after a rally (NAV increased) and less after a decline (NAV decreased).

The rebalance trades are concentrated in the **last 15-30 minutes** of the trading session (3:30-4:00 PM ET). The swap counterparty banks execute via futures, ETFs, or stock baskets at or near the market-on-close (MOC). This creates a predictable, mechanical flow in the final minutes that can be estimated in advance.

20.3 Fund Universe

The dashboard tracks 16 levered ETFs across 6 underlying indices:

Underlying	Bull Funds	Bear Funds	Total AUM (approx)
QQQ (Nasdaq 100)	TQQQ (3x), QLD (2x)	SQQQ (-3x)	~\$50B
SPY (S&P; 500)	UPRO (3x), SPXL (3x), SSO (2x)	SPXS (-3x), SDS (-2x), SH (-1x)	~\$29B
SMH (Semiconductors)	SOXL (3x), USD (2x)	SOXS (-3x)	~\$25B
IWM (Russell 2000)	TNA (3x)	TZA (-3x)	~\$2.5B
XLK (Technology)	TECL (3x)	—	~\$5.6B
VIX	UVXY (1.5x)	—	~\$0.5B

AUM figures as of May 2026. Source: Bloomberg, ETFdb. Total US levered ETF AUM ~\$203B per Citadel Securities GMI (May 2026).

20.4 Data Input: Image Paste OCR

The primary workflow for inputting AUM data is **image paste from Bloomberg**. The tab includes a drop zone that accepts Bloomberg terminal screenshots via:

- Ctrl+V (paste from clipboard)
- Drag and drop an image file
- Browse button (file picker)

When a Bloomberg screenshot is pasted, it is compressed to JPEG (max 1200px wide), sent to a Claude API proxy function (netlify/functions/ocr.mjs), and Claude extracts three data columns:

- **Ticker** — the ETF symbol (strips 'US' suffix automatically)
- **AUM** (Tot Assets column) — in \$M, used directly in the rebalance formula
- **%1D** — the daily percentage change, used for both the underlying ΔS and per-fund display

Image-extracted data supersedes all other sources. The AUM values are **persisted to the server** (Netlify Blobs) so they survive page refreshes throughout the trading day. The %1D values update from live Polygon quotes on each refresh.

Data Priority (Three Layers)

- **Layer 1 — Snapshot data:** Baseline from the 15-minute polling cycle
- **Layer 2 — Polygon quotes:** Fresh quotes fetched on tab open / Refresh for SPY, QQQ, SMH, IWM, XLK
- **Layer 3 — Image paste (BBG):** Overrides everything. Shown with ✓ (AUM) and BBG (underlying ΔS) badges

Text Paste Fallback

A text paste option is available for manual entry in format 'TQQQ 36776.10' per line. The parser handles \$, B/M/K suffixes, and various delimiters. Manual text values also override auto-fetched data.

20.5 GEX Confluence Badges

Each underlying's rebalance section displays a confluence badge that combines the rebalance direction with the options GEX regime from the Scanner/Cross-Asset tabs:

Badge	GEX Regime	Rebalance Dir.	Interpretation
■ Amplification Risk	Negative γ	Sell	Most dangerous. Forced selling into a market where dealer hedging is pro-cyclical. Close can get significantly worse than intraday low.
Momentum Reinforcement	Positive γ	Buy	Buying extends trend but absorbed by dealer counter-trend hedging. Expect grind higher, not spike.
Dampened	Positive γ	Sell	Dealer support partially offsets mechanical selling. Orderly close despite sell flow.
Reflexive	Negative γ	Buy	Potential squeeze. Dealer hedging chases rally higher while mechanical buying adds pressure.
Minimal	Any	< \$50M	Flow too small to create meaningful price impact.

All badges are clickable — tap to expand a detailed explanation of the current state, including the underlying's daily move, gamma regime, and estimated rebalance amount.

20.6 Historical Capture

The batch capture function (capture-background.mjs) automatically records levered ETF rebalance data in the daily snapshot blob after the final wave completes. The captured data includes:

- Total rebalance across all underlyings
- Per-underlying breakdown: ΔS , GEX regime, total rebalance, total AUM
- Per-fund detail: ticker, leverage, gamma, AUM, rebalance amount

The 'Show History' button on the Rebalance tab fetches the last 60 days of daily rebalance data and displays a table with columns: Date, Total, QQQ, SPY, SMH, IWM. Days exceeding \$2B total rebalance are flagged with ■ and a subtle red background. Each underlying column shows the rebalance direction and GEX regime ($-\gamma$ or $+\gamma$) that was active.

The snapshot endpoint serves historical data at: `GET /.netlify/functions/snapshot?rebalance-history=true`

20.7 Clickable Rich-Text Explainers

The Rebalance tab includes six clickable explainer targets, matching the dashed-underline pattern used throughout the dashboard:

- **Formula header** ('Rebalance = $\gamma \times \Delta S \times \text{AUM}$ ') — Full derivation with variable definitions and sourcing
- **Headline label** ('Est. Closing Rebalance') — What the headline BUY/SELL number means in practice
- **Timing note** ('Hits into close ~3:30-4:00 ET') — Why the close, how banks execute, frontrunning dynamic
- **γ column header** — The gamma table showing why inverse ETFs are disproportionately destabilizing
- **Confluence badges** — Per-underlying GEX x rebalance interaction with current state context
- **Methodology footer** — Full methodology, AUM sourcing, limitations, and source citations

20.8 Architecture

The Rebalance tab adds four components to the codebase:

- **Rebalance component** (index.html) — Frontend React component with image paste, OCR, formula engine, history display, and explainer UI
- **ocr.mjs** (netlify/functions/) — Proxy function routing image data to Claude Sonnet API for OCR extraction
- **snapshot.mjs updates** — Added save-aum/load-aum endpoints for AUM persistence, and rebalance-history query for time series data

- **capture-background.mjs updates** — Added levered ETF rebalance capture after final wave, fetching Polygon reference data for all 16 funds and computing per-fund and per-underlying rebalance amounts

20.9 Sources

The levered ETF rebalance framework draws on:

- Abdelmessih, K. — 'The Gamma of Levered ETFs' (2022). Core formula derivation and mechanism explanation.
- Constan, A. (@dampedspring) — Practical application: AUM + daily change + formula, sum across funds per underlying (2026).
- Citadel Securities GMI (Rubner, S.) — 'Flow Fragility' (May 18, 2026). Record levered ETF AUM ~\$203B, ~92% equities, ~70% tech. Institutional participation confirmed.
- Goldman Sachs — 'Rise and Reach of Retail Trading' (May 13, 2026). Retail ~25% of 3x levered ETF volume.

21. Interactive Features

Every metric label on the ticker detail card is clickable. Tap to expand the definition, current reading interpretation, and similar historical dates. This applies to all metric cards across all sections.

- **Label click** (■ icon) — Opens the metric definition panel with technical explanation
- **Value click** — Opens the current reading interpretation with percentile and similar date context
- **Chart icon** (■■) — Opens the historical time series chart for that metric across captured dates
- **CSV export** — Downloads all historical readings plus today's live data for the selected ticker

The Rebalance tab extends this pattern with its own set of clickable explainers (Section 20.7).

22. Metric Reference

Complete metric definitions for all dashboard fields. All metrics are computed from Polygon.io options chain snapshots unless noted otherwise.

Metric	Definition	Tab(s)
Net GEX	Total net gamma exposure in dollars. Positive = dealers long gamma (suppressing). Negative = short gamma (amplifying).	Scanner, Detail
Gamma Regime	Positive or negative based on Net GEX sign. Defines whether dealer hedging is counter-trend or pro-trend.	All
Flip / HVL	Strike where Net GEX changes sign. Price level where dealer behavior inverts.	Detail
Call Wall	Strike with highest call OI. Acts as resistance via dealer delta hedging.	Detail, Walls
Put Wall	Strike with highest put OI. Acts as support via dealer delta hedging.	Detail, Walls
Max Pain	Strike that minimizes total option holder P&L.; Gravitational center for OPEX pinning.	Detail
Δ P/C	Delta-weighted put/call ratio. Weights each contract by delta exposure. < 0.7 = bullish, > 1.3 = bearish.	Detail, Cross-Asset
NDF (Net Delta Flow)	Aggregate directional exposure from all options in equivalent shares.	Detail
25 Δ Risk Reversal	25-delta call IV minus 25-delta put IV. Contrarian signal at extremes.	Detail, Header
Squeeze Score	Composite 0-100 from neg gamma, flip proximity, put wall proximity, steep skew, high P/C, forward skew.	Detail, Cross-Asset
Skew Z-Score	Standardized measure of skew steepness relative to recent history.	Detail
Term Structure Slope	Near-term vs long-term IV relationship.	Detail
ODTE %	Percentage of total options volume in same-day expiry contracts.	Detail
Vol Surge	Options volume as multiple of trailing average. > 2x = unusual.	Vol Surge
Rebalance	Levered ETF mechanical flow: $\gamma \times \Delta S \times AUM$. Always in direction of move.	Rebalance
Confluence Badge	GEX regime x rebalance direction interaction. 5 states from Amplification Risk to Minimal.	Rebalance

23. Signal Effectiveness Rankings

Each signal is ranked 1-10 on standalone reliability. Most signals improve by 1-2 points when used in confluence with the GEX regime reading.

23.1 Signal Rankings

Signal	Score	Tier	Key Insight
GEX / Gamma Exposure	9/10	Primary	Regime defines how every other signal behaves. Not directional per se, but the prerequisite.
Put/Call Wall Proximity	8/10	Primary	Strongest when single strike holds 20%+ OI. Combined with regime for direction.
25-Delta Risk Reversal	8/10	Primary	Extreme readings reliably contrarian. < -8 = bullish snapback. > -1 = bearish.
Levered ETF Rebalance	7/10	Primary	Non-discretionary, predictable, concentrated in final 30 minutes. \$5-10B+ on 2% days.
Spot-Up / Vol-Up	7/10	Secondary	FOMO detector. Breaking normal spot-vol correlation signals late-stage rally.
ODTE Flow Concentration	7/10	Secondary	Enormous near-expiry gamma creates predictable intraday dynamics.
Skew Term Structure Spread	6/10	Secondary	Near-term steepening > +3pts = genuine risk-off. Best with GEX confirmation.
Options Volume Surge	6/10	Secondary	Attention flag. Tells you where to look, not what to do.
Skew Curve Classification	5/10	Tertiary	Forward skew is interesting. Becomes 7-8 when combined with neg gamma.
Put/Call Ratio	5/10	Tertiary	Classic but noisy. Better as 5-day MA.
OTM Divergence	5/10	Tertiary	Weakest vol divergence type. Hard to distinguish abnormal from steep-but-normal.
Smile Ratio	4/10	Tertiary	Simplified risk reversal. Redundant with 25Δ RR.
Cross-Asset Squeeze Score	4/10	Tertiary	Useful for filtering watchlist. Weighted average of other signals.
Spot-Down / Vol-Down	3/10	Tertiary	Highest false positive rate. Usually benign.

Levered ETF Rebalance is new to the rankings in v4. Scored 7/10 as Primary tier based on its mechanical causality, predictability, and documented market impact. The signal is most powerful in confluence with negative gamma regimes.

23.2 Signal Confluence: When Signals Combine

The highest-conviction setups occur when multiple signals align:

Combination	Score	Interpretation
Neg Gamma + Put Wall Breaking + Near-Term Skew Spike	9-10	Cascading selloff likely. Dealers forced to sell, no floor, fear concentrated.
Neg Gamma + Levered ETF Sell + Amplification Risk badge	9-10	Maximum fragility. Mechanical selling into a market where dealer hedging is also pro-cyclical. The close can significantly exceed the intraday low.

Combination	Score	Interpretation
Pos Gamma + Call Wall Approach + Extreme Negative RR	8-9	Contrarian bounce setup. Gamma pins price, everyone already hedged.
Spot-Up/Vol-Up + Forward Skew + Volume Surge	8	Speculative blowoff forming. Call chasing driving both price and IV.
Neg Gamma + Levered ETF Buy + Reflexive badge	8	Reflexive squeeze. Dealer hedging chases rally while mechanical buying adds pressure.
Neg Gamma + Spot-Down/Vol-Down	6-7	One of the few cases where the 3/10 signal matters. IV compression into weakness in negative gamma = complacency before acceleration.
Volume Surge + No Other Signals	3-4	Volume alone is noise. Without directional confirmation, volume just means activity.

23.3 Key Takeaways

1. **GEX regime** is the lens through which every other signal should be interpreted. A signal that is bullish in positive gamma may be bearish in negative gamma.
2. **Wall proximity, 25-delta risk reversal, and levered ETF rebalance** are the highest-conviction actionable signals. Each has clear mechanical causality and documented predictive power.
3. **Volume surge and spot-up/vol-up** are attention flags. They tell you where to look, not what to do.
4. **Levered ETF rebalance** adds a new dimension: known, non-discretionary flow that hits in the final 30 minutes. The direction and approximate magnitude can be estimated hours in advance. Combined with GEX regime, it identifies the most dangerous market states (Amplification Risk) and the most extended (Reflexive).
5. **Confluence is the force multiplier.** A 6/10 signal that aligns with a 9/10 regime reading and an 8/10 wall signal becomes a high-conviction setup. A 9/10 signal that contradicts other readings is less reliable than its score suggests.